

RTN analysis with FHMM as a Tool for Multi-Trap Characterization in HfO_x RRAM

Francesco Maria Puglisi, Paolo Pavan
Dipartimento di Ingegneria “Enzo Ferrari”
Università degli Studi di Modena e Reggio Emilia
Via Vignolese 905/A, 41125 Modena (MO), Italy
francescomaria.puglisi@unimore.it

Abstract—This paper presents a new technique to analyze the characteristics of multi-level random telegraph noise (RTN) in HfO_x RRAM. RTN is characterized by abrupt switching of either the current or the voltage between discrete values as a result of trapping/de-trapping activity while reading the RRAM cell. RTN statistical properties are deduced exploiting a factorial hidden Markov model (FHMM). The proposed method considers the measured multi-level RTN as a superposition of many two-levels RTN, each represented by a Markov chain and associated to a single trap, and it is used to retrieve the statistical properties of each chain. These properties (i.e. dwell times and amplitude) are directly related to physical properties of each trap.

Keywords—RTN; Multi-level; FHMM; RRAM; Trap; Noise

I. INTRODUCTION

Random telegraph noise (RTN) is usually found in MOSFETs and in other novel devices (e.g. RRAM [1]) as an abrupt change of either the voltage or the current between discrete levels. This can result in unpredictable deviation of key parameters (e.g. threshold voltage of MOSFETs) from their expected values. Currently, RTN is becoming a key issue for reliability since its effects are expected to be severe in nano-devices. Even though the physical mechanisms leading to RTN in nano-devices have not been completely assessed, it is commonly accepted that it is the result of capture and emission processes of charge carriers in/from active traps [2]. Fig. 1 shows our interpretation of the mechanism leading to both two-levels (a) and multi-level (b) RTS in metal-oxide-based RRAMs in High Resistance State (HRS) along with experimental time series [3,4]. Our previous works [3,4] exploited the color-coded time-lag plots to investigate the nature of RTN in RRAMs, showing that multi-level RTN can be seen as a superposition of many two-levels RTN. Formerly, we used hidden Markov model [5] (HMM) to investigate RTN. However, HMM can be used to extract the discrete levels in the current but cannot be used directly to extract the amplitude of each 2-levels signal in multi-level RTN. In this paper, we propose a more refined implementation of the HMM which can properly solve for the statistical properties of multi-level RTN caused by multiple traps. Traps parameters are estimated using a factorial hidden Markov model [6] (FHMM) approach: the proposed method is self-consistent (the number of active traps is automatically determined) and its implementation can be parallelized, leading to better performances w.r.t. other methods (e.g. Gibbs sampling) [7].

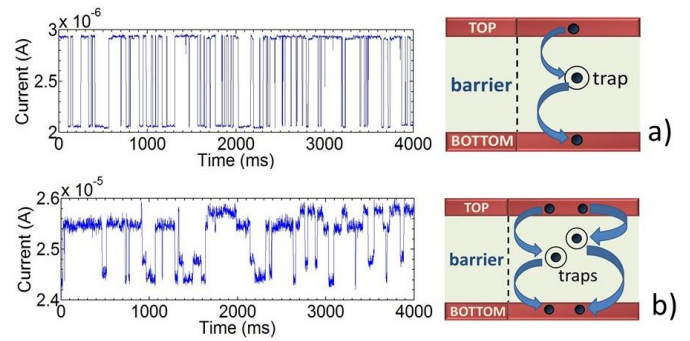


Fig. 1. a) Experimental two-levels RTN and its simplified physical mechanism involving one trap only. b) Experimental multi-level RTN and its simplified physical mechanism involving two (many) traps. Black spots are the charge carriers and the black holes are active traps. Metal-oxide-based RRAM conduction in HRS is modelled as trap-assisted tunneling via the traps in the barrier, leading to 2-levels (a) or multi-level RTN (b).

II. FACTORIAL HIDDEN MARKOV MODEL

A. Finding the Amplitudes

In the HMM [5] the input signal is modelled as a Markov chain with K hidden states (i.e. discrete levels) buried in noise. Whereas this approach is suitable to analyze a 2-levels RTN (produced by a single trap), it is inappropriate when the RTN is generated by more than one trap. Indeed the HMM detects the discrete levels: given a 2-levels RTN, the difference between the levels allows calculating the amplitude of the fluctuation, but this is not the case for a multi-level RTN. The FHMM [6] extends the HMM potential by considering the multi-level RTN as a superposition of many two-levels Markov chains [3, 4], one for each active trap, instead of modelling it as a single Markov chain with K states. This approach allows us to decompose the multi-level RTN and to retrieve the properties of each trap contributing to the RTN. Fig. 2 shows the concept of FHMM method: S_t^m represents the state of the m -th chain at time t , while Y_t represents the output of the whole process (i.e. the expected value of the multi-level RTN) at time t . Regrettably, the number of parallel Markov chains (i.e. the number of active traps), should be estimated beforehand. However this issue is solved by feeding the algorithm a large number of expected traps: the chains related to traps which are unnecessary to match the input RTN will be given a negligible amplitude and will be discarded.

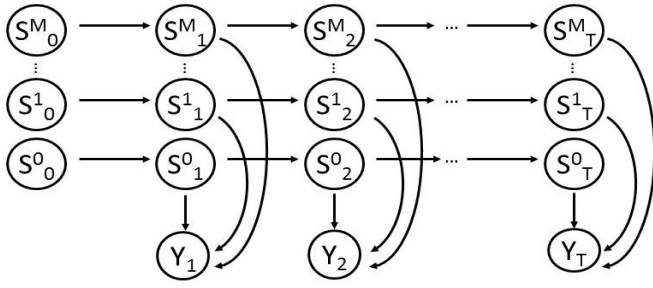


Fig. 2. Graphical representation of an FHMM. At each instant of time t , each output y_t is given by the superposition of M independent and parallel Markov chains.

B. Most likely Sequence and Dwell Times

Given the number of active traps and their amplitudes, it is possible to calculate the most probable state of each Markov chain (i.e. of each trap) in every time step t through the Viterbi algorithm [5], i.e. a dynamic programming method for finding the most likely state sequence in a HMM given a set of observations. This allows us to decompose the input signal in a superposition of different 2-levels RTN, and to extract both amplitude and capture/emission times of each chain (i.e. statistical properties of each trap).

III. RESULTS AND DISCUSSION

The proposed method has been tested using mathematically generated RTN data simulating the activity of 3 traps with additive gaussian noise. The resulting RTN is analogous to the experimental RTN traces. The number of expected traps was intentionally set to 5, a reasonably high value, in order to check the algorithm capability of automatically determining the number of active traps. Results are summarized in Tab. 1 and plotted in Fig. 3. Input data are correctly separated in three 2-levels RTN time series with remarkable accuracy. Moreover, since only 3 traps were necessary to match the input signal, the algorithm assigned a negligible amplitude to 2 out of 5 chains. The algorithm was also successfully applied to the two real experimental time series plotted in Fig.1. Results are summarized in Tab. 1 (fitted time series are not shown). Other statistical machine learning methods can solve this problem using Markov chain Monte Carlo (MCMC) approaches, the simplest of which is Gibbs sampling [7]. Although this technique is guaranteed to converge to the real probability distribution of the data, the whole space defined by unknown variables has to be sampled, resulting in an extreme computational burden as a consequence of the so-called curse of dimensionality. Moreover, Gibbs sampling technique is intrinsically non parallelizable since every step is based on the result of the previous one, preventing an efficient implementation from being possible. Conversely, the FHMM approach takes advantage of parallel computing: since the time-to-solution is strictly dependent on the initial guess of the variables, it is possible to run parallel instances of the FHMM algorithm on different cores and then choose the result maximizing the likelihood.

TABLE I. FHMM PERFORMANCES AND CHAIN CHARACTERISTICS

Trap #	Chain Amplitude [a.u.] (Generated – Fig. 3)	Chain Amplitude [a.u.] (FHMM – Fig. 3)	Chain Amplitude [A] (Fig. 1 a)	Chain Amplitude [A] (Fig. 1 b)
1	2	1.994	$8.7 \cdot 10^{-7}$	$1.0 \cdot 10^{-6}$
2	1	0.994	$1.3 \cdot 10^{-10}$	$3.2 \cdot 10^{-7}$
3	5	4.998	$5.9 \cdot 10^{-11}$	$8.6 \cdot 10^{-8}$
4	0	0.024	$1.1 \cdot 10^{-11}$	$3.4 \cdot 10^{-8}$
5	0	0.002	$1.3 \cdot 10^{-12}$	$9.6 \cdot 10^{-10}$

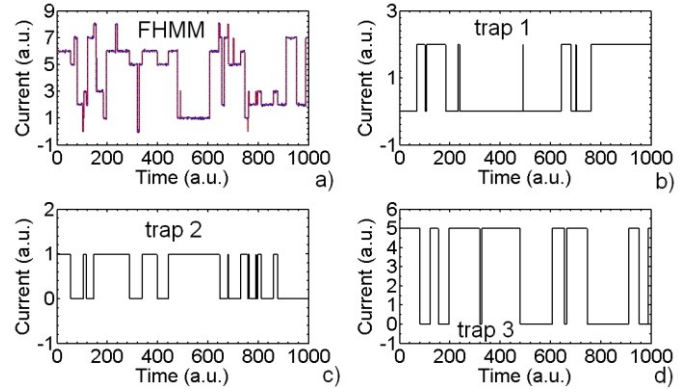


Fig. 3. a) An eight-levels RTN generated by three traps with superimposed additive gaussian noise (blue curve) and FHMM fitting (red curve). Accuracy is excellent. b, c, d) Decomposed chains amplitudes and sequences are easily found and traps characteristics can be inferred.

IV. CONCLUSIONS

In this paper FHMM was used to solve for the statistical properties of each trap contributing to multi-level RTN in HfO_x RRAM devices. A complex RTN has been separated in multiple 2-levels RTN, allowing to infer the characteristics of the traps leading to RTN. This is of crucial relevance for studying the trap-assisted conduction in novel devices. This method is comprehensive and shows excellent performance with both experimental and mathematically generated data.

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